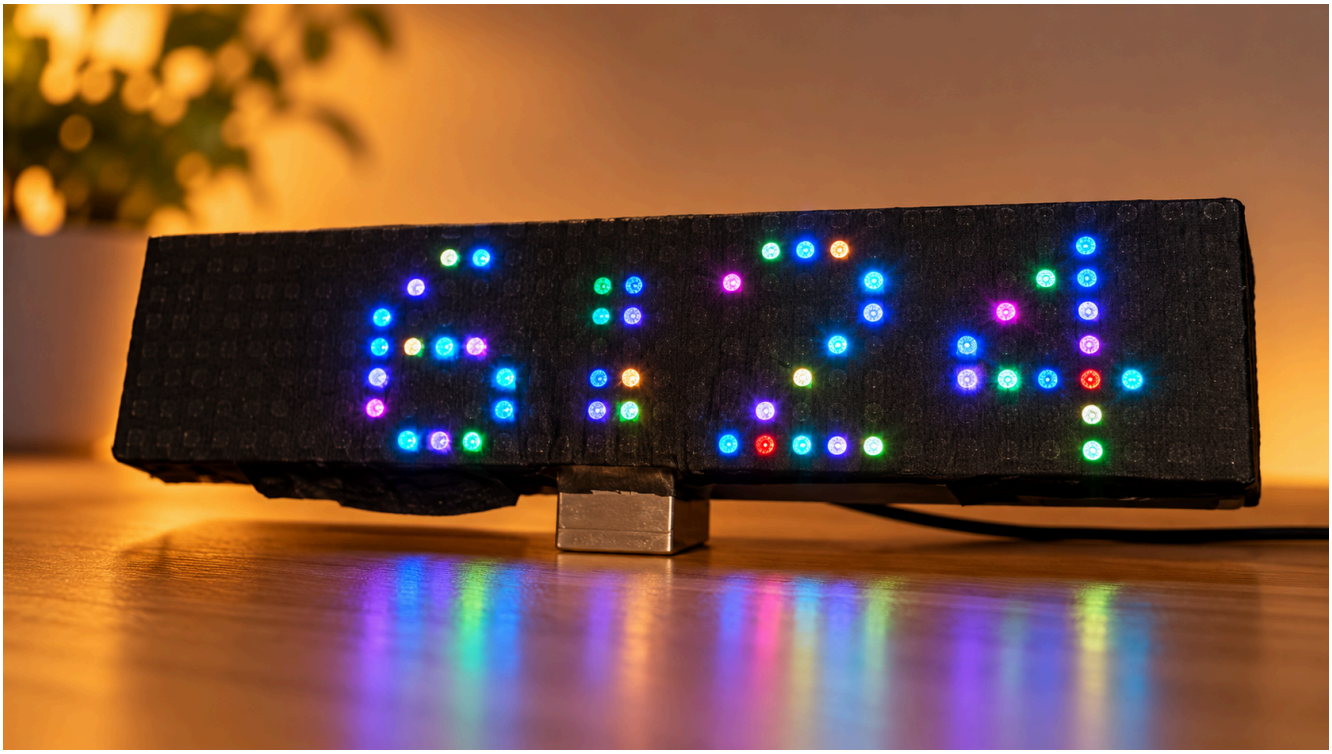
# ESP Siesta Clock with Matrix Display

*Technical Reference - English*



**A clever, super-dark clock that respects your sleep.**

## 1. System overview

The final firmware is an ESP8266 clock controller integrating a 256-pixel WS2812 display, DS3231 RTC, SSD1306 OLED, direct NEC IR decoder, EEPROM configuration, bounded WiFi/NTP sessions and a public-holiday cache. The The repository contains the final working firmware used by the device.

## 2. Architecture

| Subsystem | Role |
|---|---|
| ESP8266 | Main state machine, display, editor, networking and IR decode |
| DS3231 | Continuous time source and fallback |
| WS2812 32 x 8 | Primary time display |
| SSD1306 128 x 64 | Configuration/status UI |
| NEC IR | Primary user input |
| GPIO0 button | Boot-time WiFiManager portal trigger |
| EEPROM 128 bytes | Persistent settings and holiday cache |
| NTP | Periodic correction of RTC |
| Nager.Date endpoint | Nationwide public-holiday source when Public rule is enabled |

## 3. Pin assignment

| Signal | GPIO |
|---|---|
| WS2812 data | 13 |
| IR receiver | 14 |
| Config button | 0 |
| OLED/RTC SDA | 4 |
| OLED/RTC SCL | 5 |

## 4. Display geometry

Matrix constants: 32 columns x 8 rows, 256 WS2812 pixels. Digit X positions are 2, 8, 19 and 25; DIGIT_Y is 1. The breathing separator uses columns 15 and 16 with lit sections on rows 2-3 and 5-6 and a black row 4. Brightness range is 1-30; fresh-install day brightness is 6.

## 5. Timekeeping and WiFi lifecycle

The DS3231 is the continuous time source. On normal boot the firmware does not call WiFiManager.autoConnect(). Instead it attempts saved credentials for a bounded 20-second window. If connected, it synchronizes NTP, updates the RTC, optionally refreshes public-holiday data, then calls the WiFi shutdown path. Holding GPIO0 low at boot explicitly enters WiFiManager configuration.

The default AUTOSYNC interval is 24 hours, defined by DEFAULT_SYNC_HOURS. The interval is user-selectable from 1 to 255 hours.

## 6. Country and timezone model

Country and Time Zone are intentionally separate. The firmware exposes 249 country entries for holiday data and 63 compact timezone rules for civil-time behavior. Greece is index 0. Country selection never automatically changes timezone.

## 7. Holidays and weekend engine

Holidays has a master enable plus three independent date rules: nationwide Public Holidays, Saturdays and Sundays. A matching rule changes only the effective Night Mode end time to holidayOverrideHour:holidayOverrideMinute. The Night Mode start time and all other clock behavior are unchanged. Fresh-install override time is 08:00.

Public holiday data is cached as a 366-day bitmap occupying 46 bytes in EEPROM. Weekend-only operation is local and requires no Internet access. When Public Holidays is disabled, the holiday engine does not need to refresh online public-holiday data for Saturday/Sunday matching.

## 8. Night Mode priority

Night Mode is the brightness authority. checkBrightnessMode() computes the effective end time, including a holiday/weekend override when applicable, then enforces BRIGHTNESS_MIN. Purple editor override may change

color but is not allowed to raise brightness. This is the implementation behind the project message: “A clever, super-dark clock that respects your sleep.”

## 9. Seasonal Siesta engine

| Season | Day | Night | Calendar |
|---|---|---|---|
| Summer | 15:00-17:30 | 23:00-07:00 | 01/04-30/09 |
| Winter | 15:30-17:30 | 22:00-07:30 | 01/10-31/03 |

The editor copies the saved seasonal structures into temporary edit structures and commits only after all eight steps validate. This prevents partially edited schedules from being written to EEPROM.

## 10. OLED editor

The EditorPage enum contains: MAIN, COUNTRY, TIMEZONE, TIME, HOUR_FORMAT, FONT, SIESTA, NIGHT, HOLIDAYS, HOLIDAY_OPTIONS, DIMMER, SYNC, MONO_COLOR, MULTI_COLOR and LANGUAGE. The main menu presents 14 user-facing entries. English and Greek labels are implemented directly in firmware.

## 11. Fonts and color modes

Six matrix font styles are available. The DayColorMode enum provides MONO, SOLID, COLORFUL, FUNNY and DYNAMIC behavior. Mono mode uses a 16-color palette. Orange is reserved for Siesta indication and intentionally excluded from the user mono palette.

## 12. EEPROM map

| Address | Content |
|---|---|
| 0-3 | Night Mode start/end |
| 4 | Day brightness |
| 5 | Legacy timezone byte |
| 6-9 | Reserved legacy Siesta bytes |
| 10 | Mono color |
| 11 | AUTOSYNC hours |
| 12-14 | Unused |
| 15-16 | Day color mode + magic |
| 17-41 | Seasonal Siesta settings + magic |
| 42-44 | 12/24, font, display magic |
| 45-46 | OLED language + magic |
| 47 | Country index |
| 48 | Holidays master |
| 49-50 | Holiday override HH:MM |
| 51 | Holiday settings magic |
| 52-55 | Holiday cache metadata |
| 56-101 | 366-day holiday bitmap |
| 102-104 | Extended timezone index + magic |
| 105 | Holiday rule bit mask: Public/Saturday/Sunday |
| 106 | Holiday-rule magic |
| 107-127 | Unused |

## 13. Dependencies

- ESP8266WiFi.h
- ESP8266HTTPClient.h
- WiFiClientSecureBearSSL.h
- TZ.h
- WiFiManager.h
- NTPClient.h
- WiFiUdp.h
- Adafruit_NeoPixel.h
- EEPROM.h
- U8g2lib.h
- RTClib.h

## 14. Network and security note

The firmware uses WiFi only in bounded sessions. Public-holiday HTTPS retrieval uses the ESP8266 BearSSL client. In this build, the holiday download path uses an insecure certificate-validation mode for compatibility; transport remains encrypted, but the firmware does not authenticate the remote server certificate. This is an implementation fact to consider if the project is adapted for a higher-security environment.

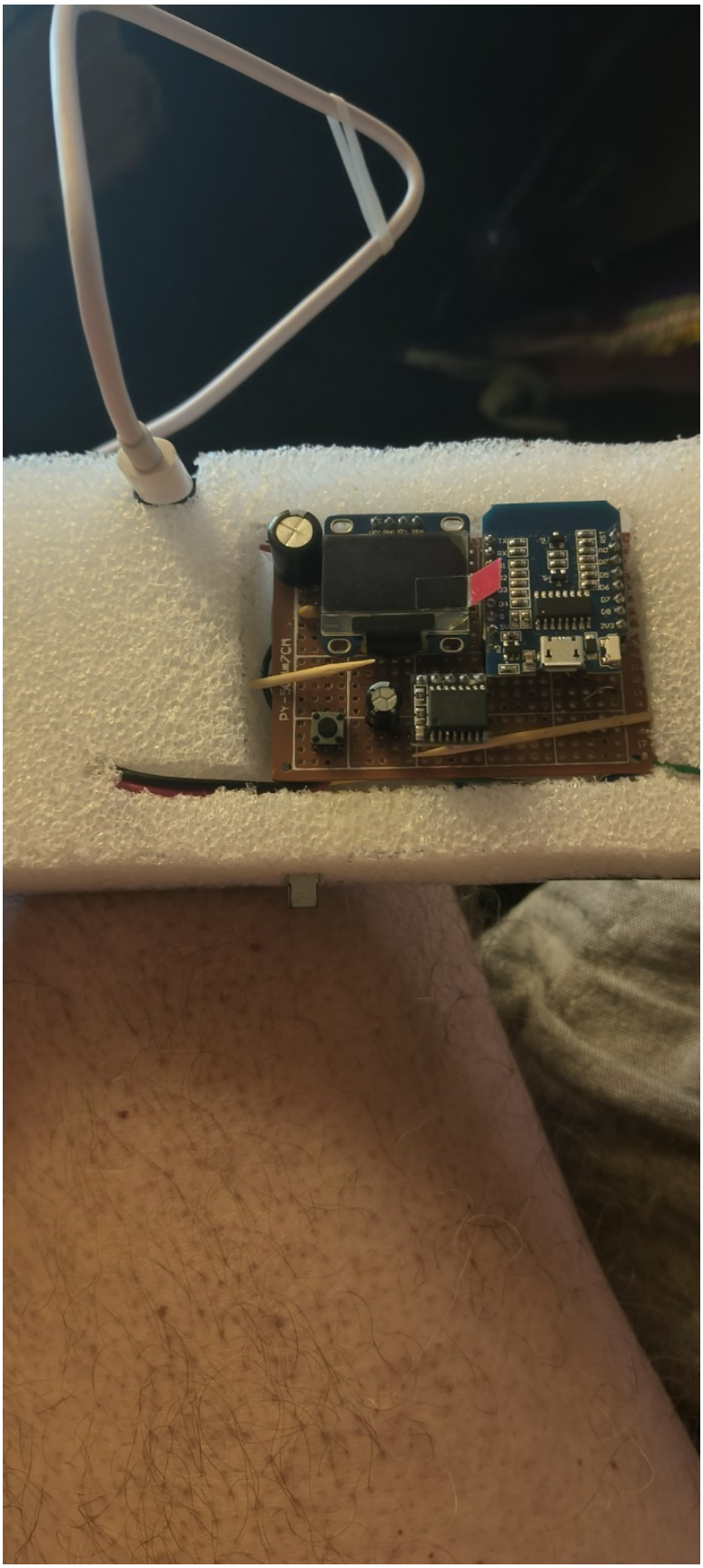
## 15. Source identity

Publication filename: firmware/ESP_Siesta_Clock_with_Matrix_Display.ino
Lines: 7,979

Bytes: 177,326

SHA-256: 8e842bcc9da1b2affbb206ff4803113f7d8c407f40a1ed76053045c1f0a75b74

The publication source uses the verified final firmware logic and concise operational comments.



*Rear electronics assembly in the photographed prototype.*